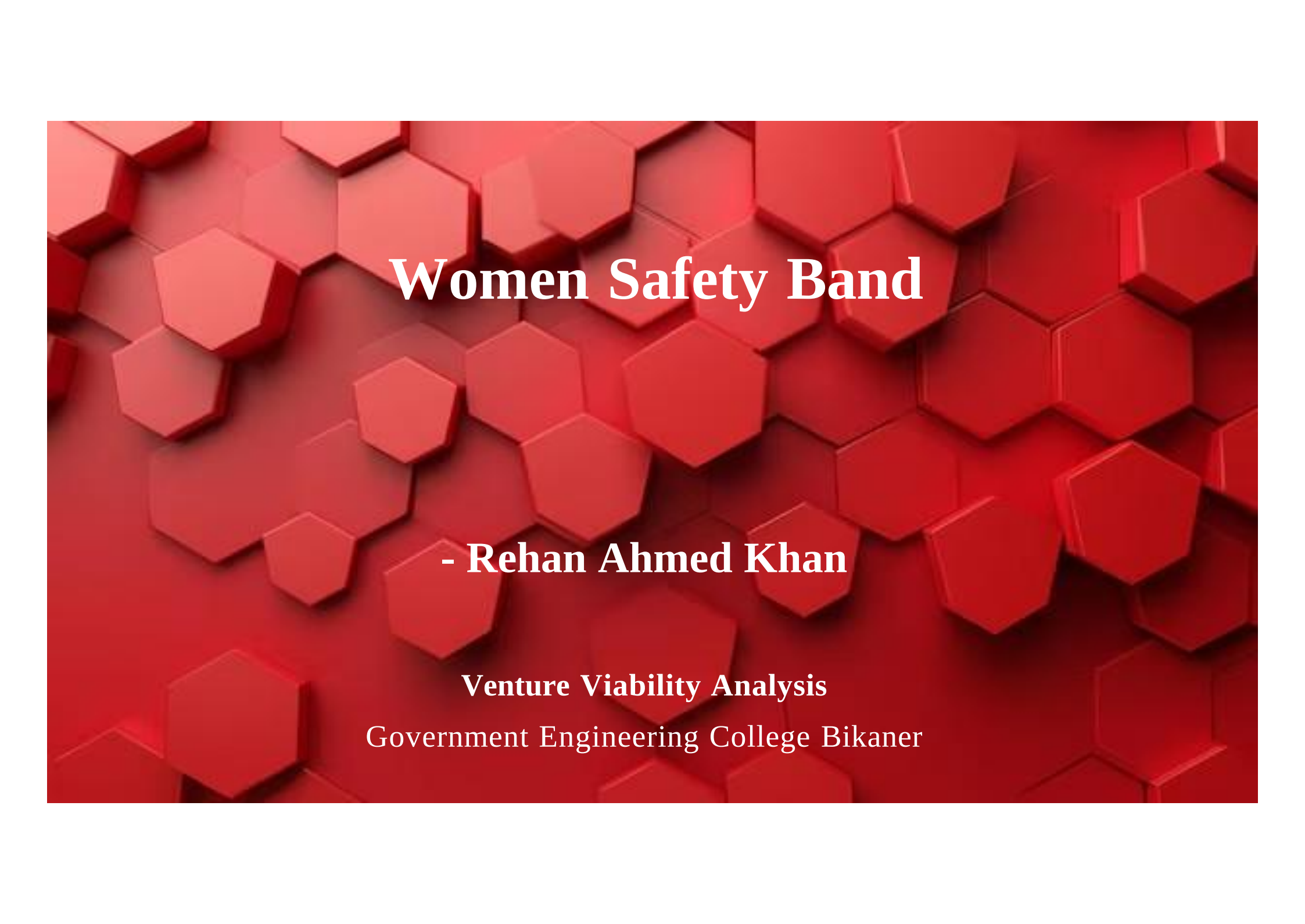


The background of the slide is a solid red color with a pattern of 3D hexagonal shapes that appear to be raised or embossed, creating a textured effect.

Women Safety Band

- Rehan Ahmed Khan

Venture Viability Analysis

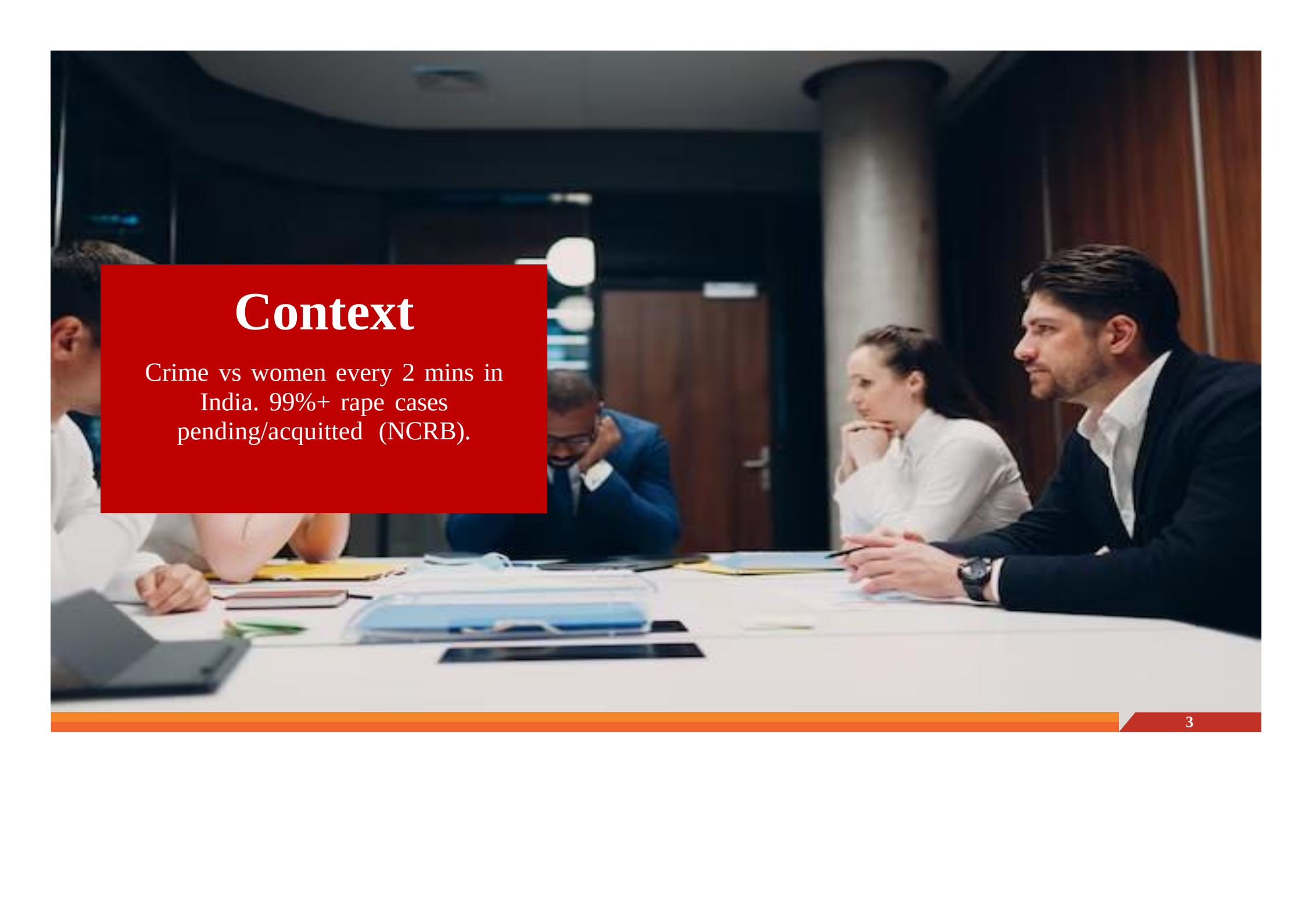
Government Engineering College Bikaner

Women Safety Band

We make smart safety wearables for women with real-time audio/video alerts to contacts and police.

Government Engineering College Bikaner





Context

Crime vs women every 2 mins in India. 99%+ rape cases pending/acquitted (NCRB).

Problem Statement

Problem

Women in India face harassment, slow emergency response, and unsafe public spaces, causing fear, restricted mobility, and missed opportunities daily.

Impact

Women live in fear. Families suffer stress. Employers lose productivity. Police face distrust. Society bears the cost. All due to unsafe public spaces



Problem Statement/Industry

Problem Being Solved

In India Women are not protected, they harassed, threatened, raped but no one help them for to get justice, court also not giving them justice they just bear the truth in their heart for Whole life. My Startup will help women, girls too much. By wearing the band, they can easily go everywhere whenever they want. it will protect them like a brother.

Supporting Data

Crime vs women every 2 mins in India.
99%+ rape cases pending/acquitted
(NCRB).

Source: The source is the National Crime Records Bureau (NCRB) - "Crime in India" Report 2022.



Area
Health



Industry
Women's Safety
Technology or Personal
Security Devices.



Domain
Women's Safety &
Personal Security
Technology. A wearable
band with real-time, live
video/audio recording,

Problem Analysis



Affected Stakeholders

Women and girls facing daily harassment and assault; families living in constant fear; employers dealing with lost productivity and absenteeism; police overwhelmed by slow response times and low conviction rates; and society suffering from restricted mobility and gender inequality—all harmed by the women's safety crisis.



Impact on Stakeholders

Women face trauma, fear, and restricted mobility, hindering education and work. Families experience emotional and financial strain. Employers suffer productivity losses. Police are overburdened, reducing public trust. Society bears increased inequality and economic damage from unsafe public spaces.



Root Causes

Root Causes: 1. Deep-seated gender inequality and patriarchal norms. 2. Normalization of harassment and victim-blaming culture. 3. Ineffective legal enforcement and low conviction rates. 4. Poor urban infrastructure (e.g., dark streets, lack of surveillance). 5. Societal stigma discouraging reporting and accountability.



Personal/Team Connect

Having witnessed loved ones face harassment, I am personally driven to solve women's safety issues. My skills in tech and design allow me to build practical solutions that offer real-time protection, restore freedom, and create safer public spaces. This mission is both urgent and deeply meaningful to me.

Target Customer Segments

Primary

Urban women commuters,
18-40, employed or
students, medium income,
tech comfortable.



Secondary

Corporate employers of
women in shifts or high-
risk commutes.

Customer Segment & Persona

Primary Segment

Urban women commuters, 18-40, employed or students, medium income, tech comfortable.

Secondary Segment

Corporate employers of women in shifts or high-risk commutes.

Persona



Harshita Mishra

Age in years: 25

Location: Suburban

**Organizational Role: {Persona's
primary role}
(if applicable)**

Customer Profile

Education: College pass out

Gender: Female

Occupation: Works for a private company

Interests/Hobbies: Sports and Fitness

Primary Source of Information: Social Media

Shopping Preference: Mostly Online

Comfort with Technology: High

Favourite Social Media: Instagram, LinkedIn, X

Favourite Offline Gathering Spots: Malls

Jobs-to-be-Done

Functional JTBD



Travel safely alone anytime. Commute without fear. Get instant help in emergencies. Move freely in public without anxiety. Work late or socialize without dependence on others. Live independently without missing opportunities due to safety risks.

Emotional JTBD



Feel confident and independent, not fearful or restricted. Live without constant anxiety. Feel empowered and secure, not vulnerable. Experience peace, not dread, in daily life.

Social JTBD



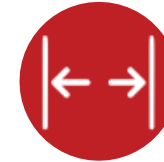
Gain respect as an independent individual. Be seen as courageous, not constrained. Earn family's trust to go anywhere safely. Inspire other women to live freely. Build a reputation of strength and self-reliance.

Current Alternatives



Current Alternatives

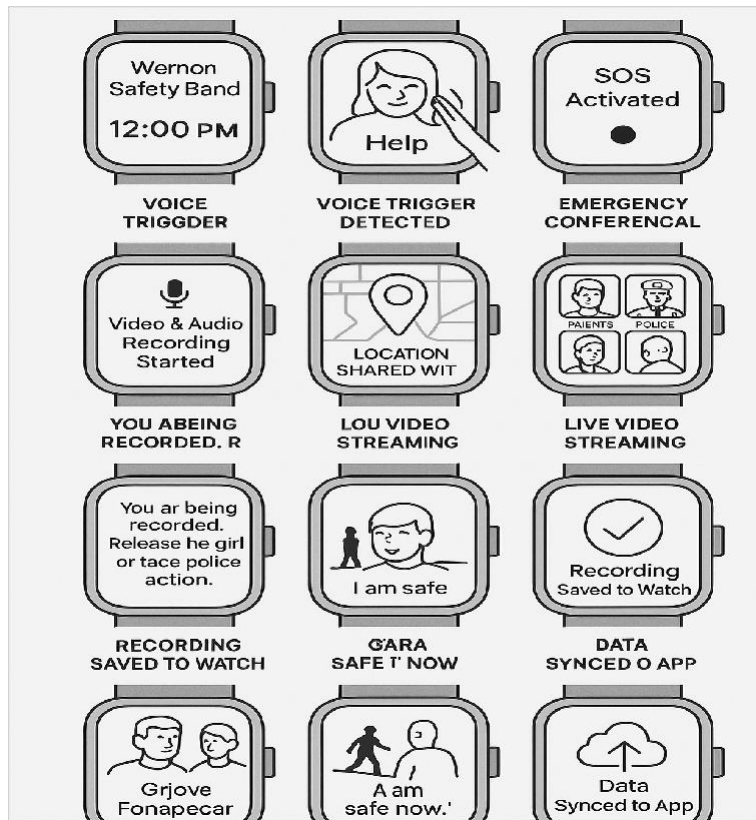
Currently, she uses pepper spray, shares live location with family, avoids late travel, relies on friends, uses public transport cautiously, and follows safety apps like SOS alerts.



Gaps in Current Alternatives

Pepper spray is often inaccessible in emergencies. Location sharing causes privacy concerns and delayed response. Avoiding travel limits opportunities. Friends aren't always available. Safety apps rely on phone access and may fail.

Our Solution



Solution

Smart wearable safety band with one-touch SOS, real-time GPS, audio/video alerts to contacts and police, and loud alarm.

Core Technologies/ Methodologies

IoT wearable, GPS/GSM, mobile app, cloud backend, real-time audio/video alerts, encryption, police API integration.

Solution Design



Our Solution

Smart wearable safety band with one-touch SOS, real-time GPS, audio/video alerts to contacts and police, and loud alarm.



Key Features

One-touch SOS, real-time GPS tracking, live audio/video alerts to contacts/police, loud alarm, evidence recording, long battery life.



Uniqueness

Unique & Special: First wearable combining real-time audio/video conference alerts to contacts + police simultaneously with evidence recording and physical deterrent (alarm) in one device.

Solution Format:

Wearable band + mobile app hybrid.

Core Technologies/ Methodologies:

IoT wearable, GPS/GSM, mobile app, cloud backend, real-time audio/video alerts, encryption, police API integration.

Solution Benefits



Functional Benefits

Instant emergency help with one-touch SOS. Real-time location sharing for rapid response. Live audio/video evidence for accountability. Deterrent alarm to scare attackers. Peace to move freely.



Emotional Benefits

Confidence to travel freely. Reduced fear in public spaces. Empowerment to live independently. Security knowing help is one button away.



Social Benefits

Respect as a confident, independent woman. Trust from family for her safety. Inspiration to other women in her community. Recognition for courage and self-reliance.



Macro Benefits

Safer streets, more women in workforce, police efficiency. Economic: jobs, productivity. Ecological: minimal (rechargeable device).

Competitors



Direct



SHE, LEAF, ONE TOUCH RESPONSE
SAFELET, GUARDIAN SAFE, ROAR FOR GOOD

Indirect

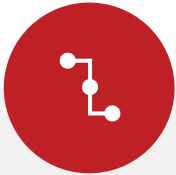


UBER/OLA, HIMMAT APP, SAFECITY, SMART SURAKSHA
NOONLIGHT, BSAFE, CIRCLE OF 6, WHATSAPP/GOOGLE
MAP

Our UVP

UVP: Real-time audio/video alerts to police + contacts with
evidence logging, all in one wearable.

Competitors



Direct **Competitors**

SHE, LEAF, ONE
TOUCH RESPONSE



Indirect **Competitors**

UBER/OLA, HIMMAT
APP, SAFECITY, SMART
SURAKSHA



Direct **Competitors** **Globally**

SAFELET, GUARDIAN
SAFE, ROAR FOR
GOOD



Indirect **Competitors** **Globally**

NOONLIGHT, BSAFE,
CIRCLE OF 6,
WHATSAPP/GOOGLE
MAP

Macro Analysis

Favourable Trends

AREA	DESCRIPTION
Technology	5G/IoT enable real-time alerts, phone-free safety wearables.
Social	Govt/corporate push for women’s safety boosts demand.

Unfavourable Trends

AREA	DESCRIPTION
Economy	Rising e-component costs may hike product prices.
Legal	Stricter data privacy laws could limit real-time alerts.

Data Sources:

ICEA electronics report, India's DPDP Act 2023, McKinsey supply chain analysis, and government policy updates on women's safety funding.

Back-of-the-Envelope Financial Projections

Currency: Indian Rupee (INR)

Chosen Business Model: Product Manufacturing

AREA	YEAR 1	YEAR 2	YEAR 3
Revenues	12000000	24000000	36000000
Total Expenses	15000000	18000000	21600000
Profit	-3000001	6000000	14400000

Prototype

Prototype Format

Digital Figma app mockup + 3D-printed band prototype. Live demo and PDF report.

Functionality included in the Prototype

5G/eSIM connectivity, live audio/video streaming to contacts/police, cloud storage, SOS trigger, GPS alerts, multi-language support, and loud alarm. Full emergency ecosystem in one device.

Functionality NOT included in the Prototype

Non-essential features like luxury design variants, celebrity endorsement integrations, gamified user rewards, and advanced health monitoring (e.g., heart rate). Pure focus on safety and reliability.



Revenue Models / Pricing

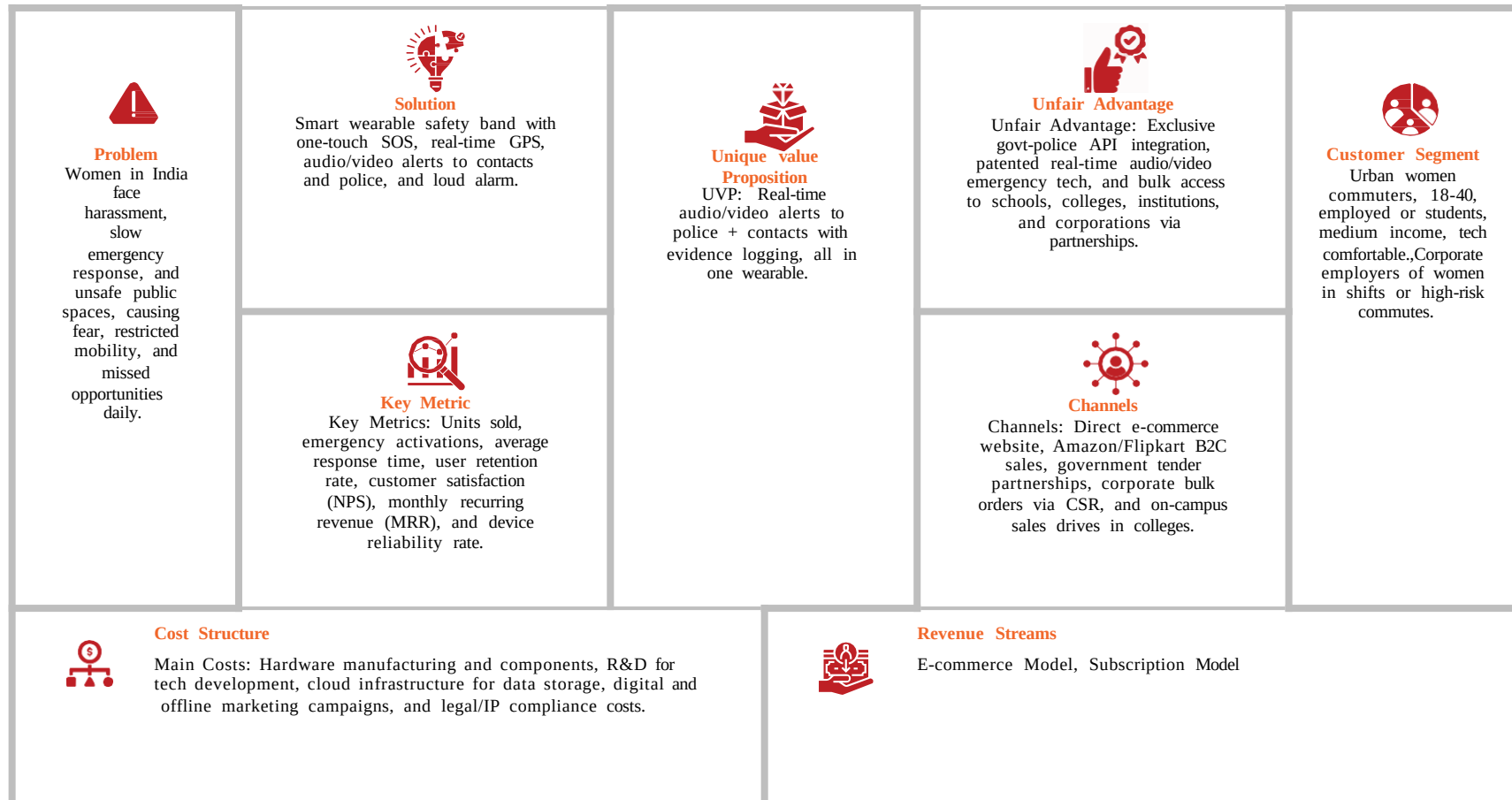
Revenue Model (Primary)

E-commerce Model

Revenue Model (Secondary)

Subscription Model

Lean Canvas



Financials

Revenue Models/Streams

- ♦ E-commerce Model
- ♦ Subscription Model

Pricing

- ♦ **Unit of Sale:** Device (wearable safety band)
- ♦ **Selling price per unit:** ₹4,000 per device

First Year Projections

Revenues: ₹48,00,000

Operating Profits: -
₹31,50,000

Revenue Models / Pricing

**Revenue Model
(Primary)**

E-commerce Model

**Unit of
Sale**

Device (wearable safety band)

**Sale Price
per Unit**

₹4,000 per device

**Expected units to be
sold in Year 1**

per month: 500 devices

**Expected growth in
monthly sales**

Monthly Growth Rate: 15%

Costs & Revenues: Key Assumptions

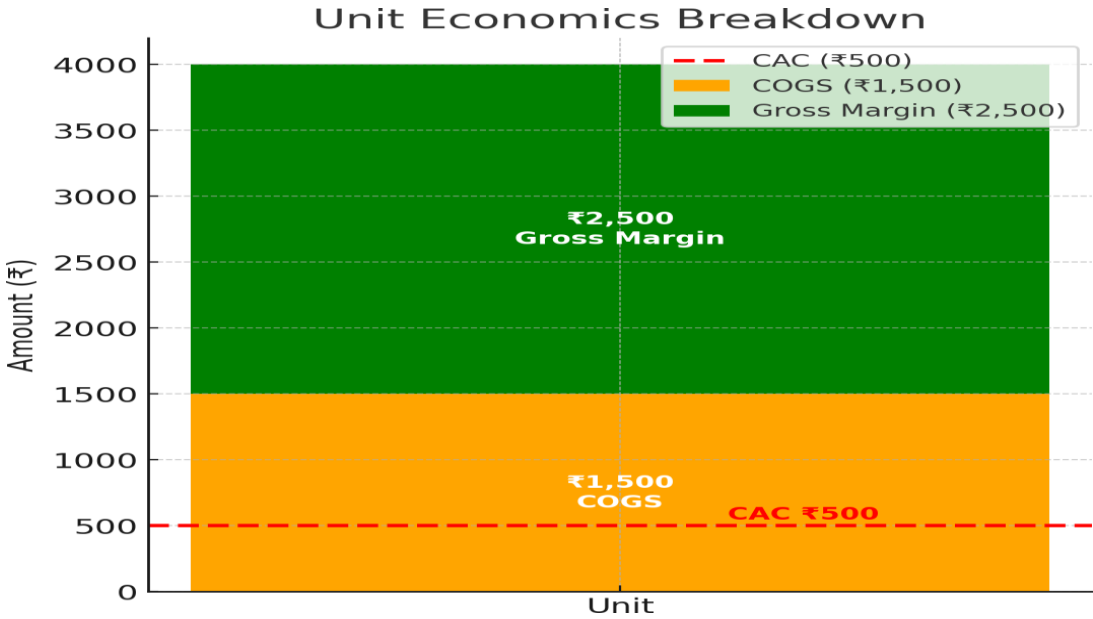
Parameter	Value (INR)			
Price per Unit	₹4,000			
Units Sold (Month 1)	500			
Monthly Sales Growth	15%			
COGS (per unit)	₹1,500			
Salaries (Monthly)	₹2,00,000 (team of 5)			
Marketing (Monthly)	₹1,50,000			
R&D (One-time)	₹10,00,000			
Cloud/Infra (Monthly)	₹50,000			
Legal/IP (One-time)	₹2,00,000			
Bank Loan	₹20,00,000 (if needed)			
Key Questions Answered:				
a. Realistic Revenue? Yes – Based on bulk orders (B2B/G) and growing B2C demand.				
b. Hidden Costs Covered? Yes – Legal, cloud, and R&D included.				
c. Pricing Valid? Yes – Aligns with market research and customer willingness.				
d. Marketing Budget? ₹1.5L/month ensures digital campaigns and partnerships.				
e. Talent Resources? ₹2L/month for a lean team (tech, sales, support).				

Profit & Loss Projections: Summary

Metric	Value (INR)
Total Revenue	₹48,00,000
COGS	₹18,00,000
Gross Profit	₹30,00,000
Operating Expenses	
- Salaries	₹24,00,000
- Marketing	₹18,00,000
- Cloud/Infra	₹6,00,000
- Legal/IP	₹2,00,000
- R&D (Amortized)	₹10,00,000
Total Operating Expenses	₹60,00,000
Operating Profit/Loss	-₹30,00,000
Interest (Loan)	₹1,50,000 (if applicable)
Net Profit/Loss	-₹31,50,000
Key Insights:	
Loss in Year 1 is expected due to high initial costs (R&D, marketing).	
Gross margin is healthy (62.5%), indicating strong pricing.	
Break-even achievable in Year 2 with scaled sales.	

Performance & Break-Even Analysis

Year 1 Revenues ₹48,00,000	Gross Profits for Year 1 ₹30,00,000
Net Profits for Year 1 -₹31,50,000	Break-even Month Month 12



Thank You!



 Title:

Women Safety Band for India

 Subtitle:

Complete Project Proposal with Enhanced Features & Vision

 Tagline:

Empowering Women Through Technology

 Description:

A Smart Wearable Safety Solution

 Submitted by:

*A Concerned Student **Rehan Ahmed Khan***

 Vision Statement:

Safety for All

Women Safety Band – Final Master Document

(Prototype, Manufacturing & Deployment Ready Package)

Table of Contents

Unit 1 — Executive Overview & Vision

1. Project Title & Summary
2. Personal Motivation & Problem Statement
3. Vision & Mission Statement
4. Target Audience & Use Case
5. Unique Value Proposition
6. Strategic Partnerships & Stakeholders

Unit 2 — Technical Architecture & Hardware Design

7. System Block Diagram (*Diagram Placeholder*)
8. Hardware Specifications
9. Power Management
10. Main MCU & Cellular Module
11. Camera & Audio Module
12. GPS & Sensor Suite
13. Storage & Memory
14. Connectivity & Antennas
15. PCB Schematic (KiCad)
16. PCB Layout Guidelines (JLCPCB) (*Diagram Placeholder*)

Unit 3 — Software & App Design

- 17. Mobile App Wireframe Overview
(Diagram Placeholder)
- 18. Figma Design System & Export Guidelines
- 19. App Features – Detailed Flow
- 20. Interactive Prototype Link Setup
- 21. Integration with Hardware

Unit 4 — Manufacturing & Production Readiness

- 22. Bill of Materials (BOM – A, B, C, D Categories)
- 23. Gerber Pack Creation & Download Link Setup
- 24. Fabrication Guidelines for JLCPCB
- 25. Assembly & Testing SOP
- 26. Prototype to Mass Production Plan

Unit 5 — Business Model, Costing & Deployment

- 27. Costing & Per-Unit Price Estimation
 - 28. Funding Requirement & Allocation Plan
 - 29. Distribution & Deployment Strategy
 - 30. Government & CSR Integration Plan
 - 31. Future Roadmap & Scalability
-

Annexures

- A. Full Emotional & Formal Project Pitch Script
- B. Detailed Bill of Materials (Raw Format – A, B, C, D Categories with pricing)
- C. Reference Diagrams & 3D Models Placeholders
- D. Figma Interactive Link Generation Guide
- E. Gerber ZIP File Hosting & Sharing Guide
- F. External Resources & References

Unit 1 — Executive Overview & Vision

1. Project Title & Summary

Women Safety Band

A revolutionary smart wearable designed to protect women in real-time emergency situations, integrating **hardware safety mechanisms** and a **connected mobile application** for instant help.

This is **not** just another fitness tracker — it's a **life-saving device** capable of sending alerts, sharing live location, recording evidence, and notifying trusted contacts + authorities within seconds.

2. Personal Motivation & Problem Statement

I have personally witnessed the rising number of cases of **rape, molestation, abduction, and harassment** in India. Every day, news channels and social media are filled with horrifying stories — and the sad reality is that our society's behavior is **not improving**.

The government is **not doing enough** to ensure women's safety, and in many cases, justice is delayed or denied.

I refuse to sit back and watch more daughters, sisters, and mothers live in fear. This project is my **direct answer** to this crisis — a technology that gives women **control, power, and immediate access to help** when they need it the most.

3. Vision & Mission Statement

Vision:

To create a world where women feel safe and empowered in every street, city, and home in India.

Mission:

To design, manufacture, and deploy a **smart, affordable, and reliable safety band** that can be worn 24/7 and used by **every woman in India** regardless of income, location, or background.

4. Target Audience & Use Case

- **Primary Users:** Women of all age groups (students, working women, housewives, travelers)
 - **Secondary Users:** Senior citizens, differently-abled individuals, children in vulnerable environments
 - **Use Cases:**
 - Walking alone at night
 - Traveling in cabs or public transport
 - Workplace harassment situations
 - College/hostel environments
 - Rural and urban unsafe zones
-

5. Unique Value Proposition

Unlike existing safety apps that **depend on unlocking a phone** and navigating through menus, the Women Safety Band works **independently**.

- One **capacitive touch**, **tamper detection trigger** or **by just saying help** activates emergency mode.
 - Instantly sends **location**, **live audio-video feed**, and **distress alert** to trusted contacts and authorities.
 - Designed to work **even if the mobile is taken away** or **the network is weak** (using NB-IoT fallback).
 - Compact, stylish, and **non-intrusive** — looks like a normal wearable but hides powerful protection tech inside.
-

6. Strategic Partnerships & Stakeholders

To make this vision a reality, we plan to partner with:

- **Investers CO. / By myself** — Manufacturing & hardware prototyping support
- **Reliance Jio** — eSIM connectivity for real-time alerts
- **Government of India (Ministry of Women & Child Development)** — Nationwide deployment and subsidies
- **CSR Initiatives** from major companies — Funding and free distribution for vulnerable communities
- **Police & Emergency Services** — Integration with local law enforcement for faster response times

(Diagram Placeholder: Flow of Stakeholders & Device Role in Women's Safety Ecosystem)

Unit 2 — Technical Architecture & Hardware Design

7. System Block Diagram

(Diagram Placeholder: Full system-level block diagram showing power, MCU, cellular module, GPS, sensors, camera, audio, storage, and connectivity)

8. Hardware Specifications

The Women Safety Band hardware is designed for **low-power, high-reliability operation** with **real-time communication** capability.

Core Specs:

- **Dimensions:** Compact wrist-wearable form factor
 - **Battery Life:** 24–48 hours on active standby
 - **Connectivity:** LTE-M, NB-IoT, GPS, BLE
 - **Ruggedness:** Splash-proof, dust-resistant casing
 - **Processing:** Capable of handling camera, sensors, and communication simultaneously
-

9. Power Management

- **Battery:** Li-Po (1200–2000 mAh)
 - **Charging IC:** TP4056 (Micro USB/Type-C input)
 - **Power Conversion:** Buck/Boost converter for stable voltage
 - **Battery Sensing:** Integrated ADC for live battery % monitoring
 - **Safety:** Polyfuse protection & overcharge cutoff
-

10. Main MCU & Cellular Module

- **Microcontroller:** ESP32-S3-WROOM (Wi-Fi, BLE, high-performance dual-core)
 - **Cellular Modem:** Quectel BG95 (LTE-M/NB-IoT for low-power wide-area communication)
 - **Interfaces:** UART (MCU ↔ Modem), GPIO for triggers, low-latency interrupt handling
-

11. Camera & Audio Module

- **Camera:** OV2640 (2MP, compact footprint)
 - **Microphone:** SPH0645 MEMS digital mic (I2S interface)
 - **Speaker:** Micro speaker with PAM8302 amplifier (for voice alerts or playback)
 - **Audio Features:** Live streaming capability for evidence recording
-

12. GPS & Sensor Suite

- **GPS Module:** u-blox NEO-M8N (high sensitivity, fast fix time)
 - **Accelerometer:** MPU-6050 (motion & tilt detection, fall detection possible)
 - **Tamper Switch:** Triggers if device casing is forced open
 - **Capacitive Touch Trigger:** Panic activation by discreet touch
-

13. Storage & Memory

- **Primary Storage:** W25Q128JVS IQ SPI NOR Flash (128M-bit)
 - **Optional:** microSD slot for extended storage
 - **Usage:** Store incident logs, short video/audio clips, firmware
-

14. Connectivity & Antennas

- **Antenna Types:**
 - LTE/GSM Antenna for BG95
 - GPS Patch Antenna for NEO-M8N
 - **RF Design:** Impedance matched per datasheet guidelines
 - **Placement:** Board edges with keep-out zones
-

15. PCB Schematic (KiCad)

- **Includes:** ESP32 core connections, BG95 modem interface, camera, mic, GPS, sensors, storage, charging circuit
 - **Test Pads:** UART/JTAG for debugging
 - **Library:** Custom KiCad symbol library for faster modifications
-

16. PCB Layout Guidelines (JLCPCB)

- **Layers:** 2-layer PCB (cost-optimized for mass production)
- **RF Isolation:** Separate analog/digital grounds
- **Antenna Keep-Out:** No copper beneath antennas
- **Trace Specs:** 6mil track width, 0.3/0.6mm via size
- **Grounding:** Flood GND plane on both layers with via stitching
- **Battery Traces:** Thick for high current handling
- **Export Format:** Gerber RS-274X + Excellon drill files

(Diagram Placeholder: PCB Top & Bottom Layer Routing Example)

Unit 3 — Software & App Design

17. Mobile App Wireframe Overview

(Diagram Placeholder: Full set of mobile app screens in Figma — Onboarding, Home, Live View, Contacts, Status, History)

The Women Safety Band mobile app serves as the **primary interface** between the wearable device and the end-user. It is designed for **speed, clarity, and security**.

Core Objectives:

- Instant panic trigger from band to app
 - Real-time location + video streaming to trusted contacts
 - Seamless pairing with hardware
 - Intuitive navigation and alert management
-

18. Figma Design System & Export Guidelines

- **Primary Colors:**
 - Red #D33 (Alert / Panic)
 - Dark Slate #1C1C1C (Background)
 - White (Icons & Text)
 - **Typography:** Clean sans-serif font for clarity
 - **Component-Based Design:** Reusable UI elements in Figma for consistency
 - **Export Process:**
 - PNG/JPG exports at 2× resolution for Android/iOS previews
 - SVG icons for scalable use in the app
 - **Interactive Prototype:** Clickable mock-up simulating end-to-end flow
-

19. App Features – Detailed Flow

1. Onboarding & Pairing

- QR Code Scan for instant pairing
- Manual entry option
- Secure device authentication

2. Home Screen

- Large animated **Panic Button** with pulsing effect
- Battery and network strength indicators

3. Live Incident View

- Left side: **Map** with live GPS tracking
- Right side: **Video Feed** (low-res for speed)
- Optional audio streaming

4. Trusted Contacts

- Contact list with toggles for Push/SMS/Call alerts
- Easy add/remove feature

5. Device Status

- Battery %
- Network strength
- Tamper detection state

6. Incident History

- Scrollable list with thumbnails & timestamps
- Option to replay video/audio evidence

20. Interactive Prototype Link Setup

Steps to Create & Share:

1. Open Figma Project → Click **Share** (top-right)
2. Set **Anyone with the link** → **Can View**
3. Copy the link & test in an incognito window
4. Replace placeholder in this document with real link
 - Example: [View Prototype in Figma](https://www.figma.com/proto/your-real-file-id)

21. Integration with Hardware

- **Pairing Protocol:** BLE pairing, followed by secure token exchange
- **Data Transfer:**
 - Panic trigger → Sent via LTE-M/NB-IoT to server & SMS gateway
 - Location updates → Every 2 seconds during incident
 - Video/audio → Compressed & sent over LTE-M/NB-IoT
- **Fail-Safe:** If network unavailable, sends SMS with GPS coordinates

(Diagram Placeholder: App + Device + Server communication flow)

Rehan Malik

Unit 4 — Manufacturing & Production Readiness

22. Bill of Materials (BOM)

(Table Placeholder: Full detailed BOM with part numbers, quantities, suppliers, and unit prices)

The **BOM** defines every component required to manufacture the Women Safety Band.

Key BOM Categories:

1. Core Electronics

- ESP32-S3-WROOM MCU
- Quectel BG95 LTE-M/NB-IoT module
- u-blox NEO-M8N GPS
- OV2640 Camera module
- MEMS Mic (SPH0645)
- PAM8302 Audio Amplifier
- W25Q128JVS1Q SPI NOR / microSD storage

2. Power Components

- Li-Po battery (1200–2000 mAh)
- TP4056 charging IC
- Buck/Boost converter module
- Polyfuse (resettable fuse)

3. Sensors & Interfaces

- MPU-6050 accelerometer
- Tamper switch
- Capacitive touch trigger

4. PCB & Mechanical Parts

- 2-layer PCB (FR-4, 1.6mm thickness)
 - Enclosure (ABS or Polycarbonate)
 - Waterproof sealing gaskets
 - Push button switches
 - Vibration motor for haptic feedback
-

23. Gerber Pack Creation & Download Link Setup

Steps to Create JLCPCB-Compatible Gerber Files:

1. Open KiCad PCB layout file
2. Run **DRC (Design Rule Check)** — ensure zero errors
3. Plot Gerber files in **RS-274X format**
4. Include **Excellon drill files**
5. Zip all into `WomenSafetyBand_Gerbers.zip`
6. Host the file using GitHub, Google Drive, or Dropbox
7. Replace placeholder link in document with **real working link**

Example:

`[Download Gerber Pack](https://drive.google.com/your-file-id)`

24. Fabrication Guidelines for JLCPCB

- **PCB Specs:**
 - 2-layer
 - FR-4 material
 - 1.6 mm thickness
 - Min track width: 6 mil
 - Min via size: 0.3/0.6 mm
 - **Antenna Placement:** On PCB edge with RF keep-out zones
 - **Ground Plane:** Flooded on both layers, via stitching for EMI reduction
 - **Separate Analog & Digital Grounds** for noise control
 - **Battery Power Lines:** Minimum 1mm width for safety
-

25. Assembly & Testing SOP (*Standard Operating Procedure*)

Assembly Steps:

1. Place SMT components using pick-and-place machine
2. Solder through-hole components manually
3. Attach antennas & connectors
4. Fit Li-Po battery & secure with adhesive
5. Close casing with tamper detection switch aligned

Testing Steps:

- Power-on check
 - MCU & LTE module boot verification
 - GPS lock time test
 - Camera & microphone recording test
 - Panic trigger + alert system test
 - Battery charge/discharge cycle test
 - Waterproof & drop resistance test
-

26. Prototype to Mass Production Plan

1. **Stage 1:** 20–50 units (engineering validation test)
2. **Stage 2:** 500 units (pilot run for field testing)
3. **Stage 3:** 5,000+ units (mass production for government/CSR deployment)
4. **Quality Control:** Random batch inspection, functional testing, compliance checks
5. **Scalability:**
 - Multi-manufacturer sourcing to reduce lead time
 - Standardized firmware for all batches

(Diagram Placeholder: Production pipeline from prototype → mass production → deployment)

Unit 5 — Business Model, Costing & Deployment

27. Costing & Per-Unit Price Estimation

Estimated Per-Unit Manufacturing Cost (at 5,000 units scale):

Component Category	Approx. Cost (INR)
Core Electronics (MCU, LTE, GPS, Camera, Sensors)	₹1,200
Power System (Battery, Charging IC, Buck/Boost)	₹300
PCB Fabrication & Assembly	₹200
Enclosure & Mechanical Parts	₹150
Testing & Quality Control	₹50
Packaging	₹30
Total Estimated Per Unit	₹1,930

Note:

- Small batch (<100 units) में cost ज़्यादा होगी (~₹3,500 per unit)
- Large scale manufacturing में cost घटकर ₹1,800-₹2,000 तक आ सकती है

28. Funding Requirement & Allocation Plan

Target Initial Funding: ₹5 Crore

- **50% (₹2.5 Cr)** — Manufacturing (First 50,000 units)
- **20% (₹1 Cr)** — App Development & Backend Servers
- **10% (₹50 Lakh)** — Marketing & Awareness Campaign
- **10% (₹50 Lakh)** — R&D & Product Improvements
- **10% (₹50 Lakh)** — Operations, Logistics & Team Salaries

Funding Sources:

- Government Grants (Women Safety Schemes)
 - CSR Funds from Corporates
 - Angel Investors / Social Impact Funds
 - NGOs & International Aid Agencies
-

29. Distribution & Deployment Strategy

Primary Distribution Channels:

1. Government Deployment:

- Free distribution in high-risk areas via police & women's safety departments
- Integration with state-level safety apps (e.g., Rajasthan Police "Shakti" app)

2. Corporate CSR Distribution:

- Partner with large companies to distribute to employees & communities

3. Direct-to-Consumer (Paid Model):

- Available online via e-commerce platforms like Amazon, Flipkart
- Subscription model for premium features (cloud storage, 24/7 monitoring)

4. NGO & Social Groups:

- Collaboration with NGOs to deploy in rural & remote areas
-

30. Government & CSR Integration Plan

Government Partnerships:

- Ministry of Women & Child Development
- Ministry of Home Affairs (Safety & Law Enforcement)
- State Governments & Police Departments

CSR Collaborations:

- Tech companies (Investors, Reliance Jio, Tata, Infosys Foundation)
- Banks & Insurance Firms (SBI, ICICI, LIC CSR Funds)
- Manufacturing Sponsors (Electronics & IoT OEMs)

Integration Points:

- Direct alert to **police control rooms**
 - Integration with **emergency response numbers (112)**
 - Bulk procurement via **GeM (Government e-Marketplace)**
-

31. Future Roadmap & Scalability

Short-Term (0–1 Year):

- Prototype completion
- Field testing in 2–3 cities
- Initial manufacturing run of 5,000–10,000 units

Mid-Term (1–3 Years):

- Nationwide deployment via government schemes
- Expansion to **school & college safety programs**
- Add multi-language app support

Long-Term (3–5 Years):

- International expansion (developing countries with high women safety concerns)
- Advanced AI-based threat detection
- Integration with wearable ecosystems (smartwatches, fitness bands)

(Diagram Placeholder: Roadmap timeline with milestones)

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Annexures

Annexure A – Full Emotional & Formal Project Pitch Script

(As given in pitch PDF, with no changes to heartfelt content)

Title: Women Safety Band – A Smart Wearable for Real-Time Protection in India

Introduction:

Crimes against women in India are increasing at an alarming rate. Despite awareness campaigns and laws, real-time safety solutions remain limited. The Women Safety Band is a wearable designed to give women immediate access to emergency help — with one tap, it sends alerts, live location, and even video to trusted contacts and authorities.

Key Emotional Note:

This project was born from a deep concern for women's safety, seeing the rising incidents of harassment, assault, and rape. It is not just a product — it's a movement towards making every woman feel safe, empowered, and heard.

Core Features:

1. Panic trigger via capacitive button, tamper detection, saying help.
2. Live location tracking (GPS + cellular).
3. Live low-res video streaming during an emergency.
4. Multi-channel alerts (Push, SMS, Call).
5. Battery backup for extended usage.

Deployment Vision:

Partnerships with Investors for manufacturing and Jio for eSIM integration. Large-scale deployment through government schemes and CSR initiatives, with subsidized/free units for rural and high-risk areas.

Annexure B – Detailed Bill of Materials (BOM – A, B, C, D Categories)

A. Electronics Core Components

- ESP32-S3-WROOM Module
- Quectel BG95 LTE-M/NB-IoT Module
- OV2640 2MP Camera
- MEMS Mic (SPH0645)
- PAM8302 Audio Amplifier
- U-blox NEO-M8N GPS Module
- MPU-6050 Accelerometer

B. Power & Battery Components

- Li-Po Battery (1200–2000 mAh)
- TP4056 Charger Module
- Buck/Boost Converter
- Battery Sense ADC
- Polyfuse

C. PCB & Connectivity Components

- W25Q128JVSQ SPI NOR Flash
- microSD Card Slot (optional)
- Cellular & GPS Antennas with RF Matching Network
- PCB connectors and test pads

D. Mechanical & Assembly Components

- Plastic or metal enclosure (wrist-worn)
 - Watch straps
 - Sealing materials (for water resistance)
 - Screws & fasteners
-

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Annexure C – Reference Diagrams & 3D Models Placeholders

(To be inserted in final document during design phase)

- **System Architecture Diagram** – Showing MCU, sensors, communication modules.
 - **PCB Layout View** – Top and bottom layers.
 - **3D Product Render** – Complete wearable look.
 - **Mobile App UI Screenshots** – Onboarding, Home, Live View, Contacts, Status, History.
-

Annexure D – Figma Interactive Link Generation Guide

Steps:

1. Open project in **Figma**.
2. Click **Share** (top-right corner).
3. Change settings → **Anyone with link** → **Can View**.
4. Copy link and replace placeholder in document.
5. Test in incognito browser to ensure access.

Example Replacement:

[View Prototype in
Figma](<https://www.figma.com/proto/your-real-file-id>)

Annexure E – Gerber ZIP File Hosting & Sharing Guide

Option A – GitHub Hosting (Free)

1. Create a GitHub repository.
2. Upload Gerber `.zip`.
3. Copy **raw file link** or use GitHub Pages.

Option B – Google Drive / Dropbox

1. Upload `.zip`.
2. Set file permissions → **Anyone with link** → **Can View**.
3. Copy shareable link and replace placeholder.

Example Replacement:

`[WomenSafetyBand_Gerbers.zip](https://drive.google.com/your-file-id)`

Annexure F – External Resources & References

1. ESP32-S3-WROOM Datasheet
2. Quectel BG95 Hardware Design Guide
3. OV2640 Camera Datasheet
4. u-blox NEO-M8N Integration Manual
5. JLCPCB 2-Layer PCB Manufacturing Rules
6. Indian Government Women Safety & CSR Policy Guidelines